SMART FACTORY PLC-LITE — VIVA VOCE PRESENTATION SLIDES

HCLTech BTech Curriculum Project (v1.0) | Candidate: BTech Learner

SLIDE 1: TITLE & PROJECT IDENTITY

Project Title: Smart Factory PLC-Lite Automation Controller
Subtitle: Edge-Autonomous Industrial IoT Controller with Cloud SCADA & ML Maintenance
Candidate: BTech Learner (Embedded Systems & IoT) | HCLTech Training Program
Live Web SCADA: https://smart-factory-plc-lite-automation-c.vercel.app
Live API Gateway: https://smart-factory-plc-lite-automation.onrender.com/health
GitHub Repo: https://github.com/vishwas789blip/Smart-Factory-PLC-Lite-Automation-Controller (v1.0)

SLIDE 2: INDUSTRY MOTIVATION & PROBLEM STATEMENT

• High Cost of Legacy PLCs: Siemens S7 & Allen-Bradley cost ■35,000–■85,000 + ■1,20,000 SCADA licenses.
• Cloud Latency Hazard: Centralized cloud controllers incur 2–5s network delays during emergency halts.
• Reactive vs Predictive: Traditional PLCs alarm only after machines break; zero bearing wear detection.
• Our Solution: Open-architecture sub-■1,900 INR controller with < 15ms autonomous edge interlocks and AI maintenance.

SLIDE 3: 4-TIER DISTRIBUTED SYSTEM ARCHITECTURE

• Tier 1 (Physical Edge): ESP32-WROOM-32, DHT22, ACS712 (Current), SW-420 (Vibe), MQ-135, Relays, OLED.
• Tier 2 (Message Broker): HiveMQ Public MQTT Broker (broker.hivemq.com:1883) with QoS 1 and LWT.
• Tier 3 (Cloud Backend): Express.js REST & WebSocket API on Render + MongoDB Atlas database.
• Tier 4 (SCADA & Intelligence): Vite React 19 SCADA Web Console on Vercel + Python Isolation Forest ML.

SLIDE 4: BILL OF MATERIALS (BOM) — ■1,840 INR UNIT COST

1. ESP32 DevKit v1: ■450 | 2. DHT22 Temp/Humidity: ■180 | 3. ACS712 Current: ■160
4. SW-420 Vibration: ■90 | 5. MQ-135 Gas: ■150 | 6. 4-Ch Relay Board: ■180
7. 0.96" I2C OLED: ■180 | 8. 5V 2.5A Power Adapter: ■250 | 9. Passives & Wires: ■200
TOTAL UNIT COST: ■1,840 INR (~$22 USD) — > 95% Cost Reduction compared to commercial PLCs!

SLIDE 5: HARDWARE PINOUT & ADC1 NOISE ISOLATION

• Critical Engineering Design: All analog sensors routed strictly to ADC1 channels:
- GPIO 32: ACS712-30A Current Sensor (ADC1_CH4)
- GPIO 34: MQ-135 Hazardous Gas Sensor (ADC1_CH6)
- GPIO 35: SW-420 Vibration Velocity Sensor (ADC1_CH7)
• Why ADC1? Prevents internal hardware conflict with ESP32 Wi-Fi RF transceiver on ADC2!

SLIDE 6: AUTONOMOUS EDGE SAFETY INTERLOCKS (< 15ms)

• Autonomous Local Control: Bypasses cloud network latency during critical hazards.
• Over-Temperature Hazard: Trips Relay 1 (Auxiliary Fan) in 8.2 ms at > 75.0 °C.
• Vibration Shock (ISO 10816 Zone D): Halts Conveyor Motor in 6.4 ms at > 4.5 mm/s to prevent bearing seizure.
• Moving Average Filter: 10-sample circular buffer attenuates electrical noise by 88%.

SLIDE 7: HCLTECH USE CASE 3: PREDICTIVE MAINTENANCE ML

• Technology: Python, Pandas, NumPy, Scikit-Learn (Isolation Forest), Streamlit.
• Feature Engineering: Vibration RMS, winding temperature, 3-phase industrial power draw.
• Anomaly Detection: Unsupervised Isolation Forest isolates mechanical bearing fatigue days before seizure.
• Export: Downloadable maintenance audit report for plant engineers.

SLIDE 8: VERIFICATION MATRIX & SUBMISSION CHECKLIST

• Monorepo Build & Typecheck: PASSED (0 errors across 6 workspaces)
• Backend Unit Tests (Vitest): 14/14 PASSED
• Python ML Tests (PyTest): 6/6 PASSED
• Realistic Stress Tests: Wi-Fi drop (3.0s reconnect), cold boot (1.4s), noise rejection (σ 4.12 -> 0.48).
• Release: Tagged v1.0 on GitHub main branch.